HowToWriteAnArduinoProgram

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Contents

(ASSUMING BASIC ENVIRIOMENT IS SET UP)

```
# include <Arduino.h>
void setup
{
  // Here goes code that would Ran on Start up
}
void loop ()
{
  // Code that will excuated while the Arduino is running
}
```

0.1 Turning builtin Led on

```
# include <Arduino.h>
void setup
{
  pinMode(LEDBUILT, OUTPUT); // set the pin 13 to be in output
}
void loop ()
{
  digitalWrite(LEDBUILTIN, HIGH); // set a low voltage
  delay(1000); // delay 1 sec
  digitalWrite(LEDBUILTIN, LOW); // send a low voltage
  delay(1000); // delay 1 sec
  // Code that will excuated while the Arduino is running
}
```

1 Circut to light a LED on

A red LED can turn with 1.2v and 5 to 20mA of current, but the Arduino works with a current of 5v and very high current. To prevent the led from recieving damage we would add a resistant to it.

1.1 Formula for this:

$$R = \frac{V_s - V_1}{I}$$

The elements on this formula are

1. V_s is the current is the suply voltage (5v)
2. V_1 is the LED forward voltage drop (1.7v)
3. I is the current required by the led (10mA) in Amp so, (0.001A)

Replacing the values on the formula we would find that

$$R = \frac{5\,\mathrm{V} - 1.7\,\mathrm{V}}{0.001\,\mathrm{A}}$$
$$R = \frac{3.3\,\mathrm{V}}{0.001\,\mathrm{A}}$$
$$R = 330\,\Omega$$

The general formula to the this is:

$$\mathrm{V} = \mathrm{I} * \mathrm{R}$$

V for voltage I for current R for resistance